2024 AUSTRALIAN SCIENCE OLYMPIAD EXAM

BIOLOGY

TO BE COMPLETED BY THE STUDENT. USE CAPITAL LETTERS.

First Name: ………………..…..………… Last Name………………………………….
Date of Birth: ……/…..../..….

□ Male □ Female □ Unspecified Year 10 □ Year 11 □ Other: …….

Name of School: …………………………………………….………………..**State:** ……...

Examiners Use Only:

2024 AUSTRALIAN SCIENCE OLYMPIAD EXAM
BIOLOGY

Time Allowed
Reading Time: 10 minutes
Examination Time: 120 minutes

INSTRUCTIONS

- ***Attempt all questions in ALL sections of this paper.***
- **Permitted materials: non-programmable, non-graphical calculator, pens, pencils, erasers and a ruler.**
- **Answer all questions on the MULTIPLE CHOICE ANSWER SHEET PROVIDED. Use a pencil.**
- **Marks will not be deducted for incorrect answers.**

Integrity of Competition

If there is evidence of collusion or other academic dishonesty, students will be disqualified. Markers' decisions are final.

MARKS

- **1 mark for each question**
- **Total marks for the paper 70 marks**

1. What is the process by which plants make their food?
    a. Respiration
    b. Photosynthesis
    c. Transpiration
    d. Fermentation

2. Which vitamin is obtained through exposure to sunlight?
    a. Vitamin A
    b. Vitamin B12
    c. Vitamin C
    d. Vitamin D

3. What is the name of the process by which water vapour becomes liquid?
    a. Evaporation
    b. Condensation
    c. Sublimation
    d. Precipitation

Questions 4 and 5 relate to the following information.

Australian animals have evolved to deal with heat in many different ways. Kangaroo species can sweat but, strangely enough, only while hopping. The rest of the time, if they get hot, they must find other ways to deal with high temperatures.

4. A kangaroo has a dense network of capillaries in its forearms. It licks its arms when it is overheating.

Source: Creator: ©Theo Allofs | Credit: ©Theo Allofs/theoallofs.com
Copyright: ©Theo Allofs.

Which type of cooling are they utilising when licking their forearms?

a. Behavioural

b. Evaporative

c. Radiation

d. Convection

5. During the heat of the day, mobs of kangaroos are found gathered under the shade of trees saving feeding and movement for dusk.

Which method of cooling are they utilising in this situation?

Source: Australian Broadcasting Corporation

a. Behavioural

b. Evaporative

c. Social

d. Convection

6. Nocturnal species are animals that are most active at night, carrying out activities such as hunting and mating.

Which of the following adaptations is **least** likely to be found in a nocturnal animal?

a. Reduced metabolism during daytime

b. Better vision in dark conditions

c. Larger ears

d. Ability to survive in extremely high temperatures

7. What is the common name for di-hydrogen monoxide?

   a. Vinegar
   b. Salt
   c. Water
   d. Alcohol

8. What is the main gas found in the air we exhale?

   a. Oxygen
   b. Nitrogen
   c. Carbon dioxide
   d. Argon

9. Which of the following statements about living cells is **false**?

   a. Most are microscopic
   b. They are found in all animals but not in all plants
   c. They are the smallest basic units that can carry out all of the functions that we normally define as life
   d. They are the basic building blocks of all living things

10. Which of the following components is found in plant cells and not in animal cells?

    a. Cytoplasm
    b. Cell membrane
    c. Cellulose
    d. Nucleus

11. What is the only continent without reptiles or snakes?

a. Europe

b. Australia

c. Antarctica

d. North America

12. Which of the following statements is **most** likely **true** abut bioluminescence in organisms?

Jervis Bay, Australia

a. It is primarily used for camouflage during the day

b. It is a byproduct of photosynthesis

c. It is a form of communication between members of the same species

d. It is only observed in nocturnal species

13. Adenosine triphosphate (ATP) is the main form of usable energy in cells. This energy is released when ATP is converted into ADP (adenosine diphosphate) by the removal of one molecule of water (a hydrolysis reaction). The molecular formula of ADP is $C_{10}H_{15}N_5O_{10}P_2$.

Given that the molecular formula of inorganic phosphate is H_3PO_4, what is the molecular formula of ATP?

a. $C_{10}H_{18}N_5O_{14}P_3$

b. $C_{10}H_{16}N_5O_{13}P_3$

c. $C_{10}H_{15}N_5O_{10}P_3$

d. $C_{10}H_{15}N_5O_{14}P_3$

14. Australia has one of the highest rates of skin cancer in the world. An estimated two in three Australians will be diagnosed with skin cancer during their lifetime.

Delivered in partnership with the Cancer Council, the Australian Government committed to a $10 million national campaign that highlighted the importance of being SunSmart to drive down rates of the country's most common, most costly and one of our most preventable cancers.

In 2022 the campaign was adopted Australia wide and is illustrated below.

Which of the below methods is best to determine the effectiveness of the campaign by?

a. surveying a subset of the Australian population, asking if they remember seeing or hearing the campaign.

b. counting the number of people on a specific day and time who are wearing hats and sunglasses.

c. measuring exposure to the sun and skin cancer incidence in a subset of the Australian population.

d. comparing skin cancer incidence before and after the campaign.

15. Koen's father, Kenny, has 5 children. The names of the oldest four are: Kaka, Keke, Kiki and Koko. What is the name of the 5th child?

a. Kuku

b. Kenny Jnr

c. Koen

d. Kuky

16. The erythrocytes (red blood cells) below were imaged using an electron microscope.
The magnification is x3000 and the ruler measures the central cell as being 2 cm in diameter.

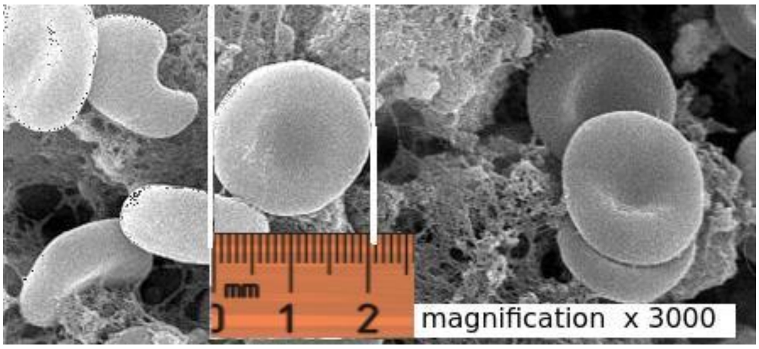


Image ref: https://thinkib.net/biology/

Estimate the actual size of this erythrocyte.

a. 6.6 µm

b. 6000 µm

c. 60 µm

d. 0.6 µm

Questions 17 to 21 relate to the following information.

Frugivorous animals play crucial roles dispersing seeds away from parental plants and influencing its ability to expand its population or colonise new habitats (plant recruitment). Frugivores thrive mostly on raw fruits or succulent fruit-like produce of plants such as roots, shoots, nuts, and seeds. The Pale-breasted Thrush (*Turdus leucomelas*) is a frugivore that swallows whole the fruits of a fleshy-fruited plant, *Miconia rubiginosa*.

Mariana Campagnoli and colleagues (2024) conducted an experiment where they aimed to answer if individual traits (body mass and sex) of *T. leucomelas* affected seed dispersal.

They tested the following hypotheses:

(a) individual traits influence seed retention time and germination,
(b) seed retention time affects seed germination.

Twenty-one birds were utilised in the experiment. Birds were categorised by sex and individually weighed. Results are displayed in Table 1.

	Body Mass (g)

Male	73.2, 71.5, 66.8, 63.4, 72.7, 65.6, 68.1, 70.9, 71.9, 67.7, 65.2, 67.6
Female	70.5, 61.3, 67.2, 64.1, 70.7, 62.8, 69.0, 66.7, 65.9

Table 1. Individual groupings of *T leucomelas* by sex and body mass (g)

17. Calculate the **mean** body mass of both males and females.

a. Males 69.1 g : Females 67.0 g

b. Males 68.7 g : Females 66.5 g

c. Males 67.2 g : Females 70.5 g

d. Males 67.7 g : Females 66.7 g

18. Find the **median** body mass of both males and females?

a. Males 65.60 g : Females 70.70 g

b. Males 72.70 g : Females 64.10 g

c. Males 67.90 g : Females 66.70 g

d. Males 66.85 g : Females 67.41 g

The researchers collected ripe fruits from five individuals of *M. rubiginosa* the day before the experiment and stored them at room temperature until the following morning. They offered five *M. rubiginosa* fruits (one from each plant individual) simultaneously, to each one of 21 T. *leucomelas* individuals through one morning (07:00–12:00 h).

Birds were monitored individually from a distance of 1.5 m, which allowed the researchers to record the exact moments of fruit ingestion and defecation.

Results of the experiment are displayed in Figure 1 and 2.

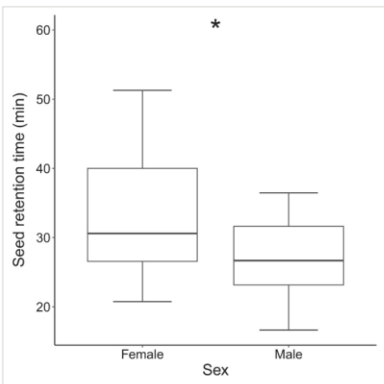


Figure 1. Boxplots showing differences in seed retention time in the gut for females (n = 9) and males (n = 12) of *Turdus leucomelas*. Whiskers represent minimum and maximum values; the box indicates the interquartile range; the horizontal bar denotes the median. The asterisk indicates significant differences in seed retention times of females and males (p = .044).

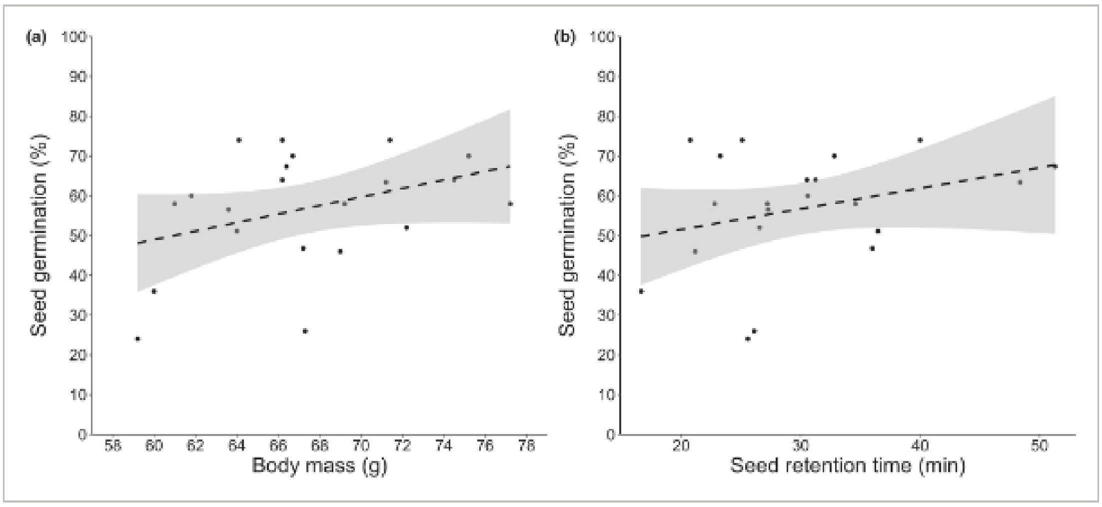


Figure 2. Scatter plots showing the relationship (dashed lines) between (a) body mass and proportion of seeds germinated, and (b) seed retention in the gut and proportion of seeds germinated. The shaded area represents the standard error of the tendency line.

19. Considering all birds, seed retention time ranged from:

    a. 20 to 38 minutes

    b. 17 to 51 minutes

    c. 17 to 38 minutes

    d. 20 to 51 minutes

20. Campagnoli and colleagues (2024) found:

    a. That *T. leucomelas* females presented higher gut retention times than males, and those seeds retained for longer periods and dispersed by heavier birds had increased gemination probability.

    b. That *T. leucomelas* males presented higher gut retention times than females, and those seeds retained for longer periods and dispersed by heavier birds had increased germination probability.

    c. That T. leucomelas females had greater body mass than males, and those seeds retained for longer periods had decreased germination probability.

    d. That *T. leucomelas* females presented higher gut retention times than males, and those seeds retained for shorter periods and dispersed by lighter female birds had increased gemination probability.

21. Based on the above data which sex (M or F) of *T leucomelas* would have a greater impact on plant recruitment?

   a. Male

   b. Female

22. *Koencooleos giganteus* is an alien plant species from the Biolympia family. It thrives in wet, cold environments, and can survive in sub-zero conditions for up to 3 weeks. Recently, scientists have been conducting extensive research on it, and are interested to see how it compares to plants from Earth.

   Which of the following best describes an example of evolution?

   a. There are no natural predators of *Koencooleos giganteus* found in regions that experience regular snowfall.

   b. Scientists choose the biggest specimens of *Koencooleos giganteus* when breeding them to maximise crop yield.

   c. The average size of *Koencooleos giganteus* plants is smaller than it used to be during the Mesozoic Era.

   d. A *Koencooleos giganteus* plant that scientists are studying grows taller in response to changing light sources.

23. Ethanol is metabolized at a constant rate of about 120 mg per hour per kg body weight, regardless of its concentration. The legal driving limit of alcohol is 160mg/100mL. Water makes up 60% of total body weight. If a 70 kg person were at twice the legal driving limit, how long would it take for their blood alcohol level to fall below the legal limit?

   a. 4 hours

   b. 6.67 hours

   c. 8 hours

   d. 10 hours

24. Anne owns a farm with 100 different chicken coops. Each coop has 20 hens, and each hen has 2 chicks. They also buy 1 rooster for every 4 hens.

    How many chickens are there on the farm?

    a. 5400

    b. 6000

    c. 6500

    d. 7200

25. What number is **X**?

4	6	16
6	3	12
16	12	**X**

    a. 15

    b. 28

    c. 35

    d. 40

Questions 26 and 27 relate to the following information.

Action potentials are electrical impulses that travel along neurons in nerve fibres, and these electrical signals are involved in initiating muscle contractions. Before the action potential begins, the inside of the cell is more negatively charged compared to the outside, due to more negative ions (e.g. Cl^-) intracellularly and more positive ions (e.g. Na^+) extracellularly. This state is called resting membrane potential. Membrane potential refers to the difference in voltage (electrical potential) between the inside and outside of the cell.

The following is a simplified explanation of the steps of an action potential. The action potential starts when a stimulus triggers an influx of Na^+ into the cell. When the membrane potential reaches a threshold value, more Na^+ channels open, allowing for further influx of Na+ into the cell. When the membrane potential reaches its peak, these Na+ channels begin to close. The next step of the action potential involves the efflux of K+ ions, causing the cell to return to its negatively charged state. Finally, an ATP-dependent Na+/K+ pump restores the initial ion concentration gradients and resting membrane potential.

26. Drug X opens additional ion channels that constantly allow Na+ to leak into neurons. If drug X is applied to a neuron, it will:

    a. Be more likely to fire an action potential.

    b. Be less likely to fire an action potential.

    c. Cause no change to the likelihood of an action potential firing.

    d. Elicit no action potential.

27. Drug Y inhibits the ATP-dependent Na^+/K^+ pump. If drug Y is applied to a neuron, the next action potential will be:

    a. Larger than normal.

    b. The same as normal

    c. Smaller than normal

    d. There will be no action potential after drug X is added

28. Human beings and most other animals will grow during the beginning of their life time but eventually in maturity they will stop growing any larger. This is called determinate growth. However, this does not apply to most plants that will undergo a constant rate of annual growth throughout their lives. This is called indeterminate growth.
    These plants are only able to undergo indeterminate growth because they possess a special type of tissue called meristems. Two types of meristems are the apical and lateral meristems. Both types are found in the roots and the stems of plants. The cells in lateral meristem tissue replicate in order to widen shoots and roots. The cells in apical meristem tissue replicate in order to extend the shoots and roots.

    Steven has a collection of several different plants that undergo indeterminate growth. He cuts 2 cm off the top of the highest stem from each of these plants. Some of the plants continue to grow just like normal, but many plants never grow taller again.

    Choose the option that best describes a reasonable explanation for this difference:

    a. Plants that stop growing taller have apical meristems in different regions of the stem to plants that keep on growing taller.

    b. The plants that stop growing are undergoing changes in the growth as a response to the stress of having their stem removed

c. The plants that continue to elongate are not eaten by herbivores, so do not have to worry about elongating a shoot that might just be eaten again.

d. The secondary meristems act more prominently than the apical meristems in the plants that do not elongate the shoot.

Questions 29 and 30 relate to the following information.

According to the widely accepted scientific model of evolution, the genetic make-up (and hence characteristics) of a group of organisms changes gradually each generation in the way that is most favourable for their survival and ability to reproduce in the current surroundings. Speciation occurs when for some reason a subgroup of these organisms have their characteristics and genetic make-up changed in a distinct way from the rest of the organisms in the group. Over time the sub-group becomes distinct enough from the still surviving original group that they can be called a different species. We call an organism viable if it is capable of survival and reproduction. The traditional notion of a species says that the subgroup will be a species if they can produce viable offspring with each other but not with any organisms outside of this subgroup. This in practice is an ambiguous statement.

29. Which of the following is **<u>least</u>** likely to cause speciation?

a. A river forms between a group of animals that they cannot cross

b. A subgroup of the original group has a sudden shift in sexual preference

c. The conditions of a group of organisms' environment becomes harsher very quickly

d. Some members of the group have a change in chromosome count that does not detriment their survival

30. Which of the following **<u>does not</u>** demonstrate a flaw in the traditional notion of a species given above:

a. There are very closely related geckos where groups of these geckos can all produce offspring with members of nearby groups but never with groups further away.

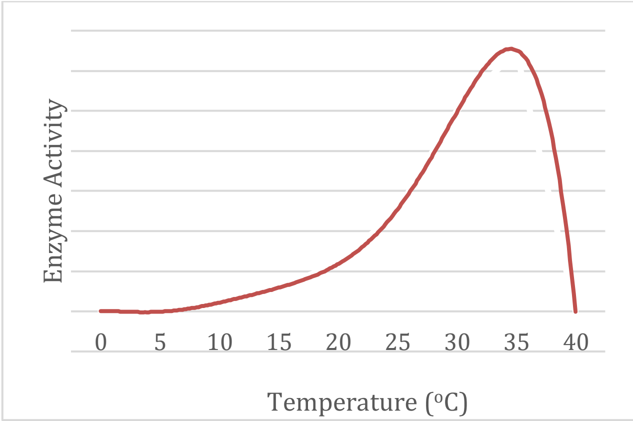
b. It is not possible to categorise extinct species based on this method

c. Some individuals in a species cannot reproduce at all

d. Bacteria reproduce asexually

Questions 31 to 34 relate to the following graph.

The graph below displays the activity of an enzyme in relation to rising temperature.



31. At which temperature is the enzyme most active?

At which

a. 25 °C

b. 30 °C

c. 35 °C

d. 40 °C

32. What is the name given to this temperature?

a. Best

b. Optimal

c. Favourable

d. Ideal

33. What happens to enzyme activity as the temperature drops below 25 °C?

a. Activity decreases

b. Activity is unchanged

c. Activity increases

d. Unable to determine activity change

34. What will happen to the activity of the enzyme if the enzyme is warmed to 45 ^{o}C then cooled to 30 ^{o}C?

a. Enzyme activity increases and stops at 40 ^{o}C. The enzyme is denatured and will not function again at 30 ^{o}C.

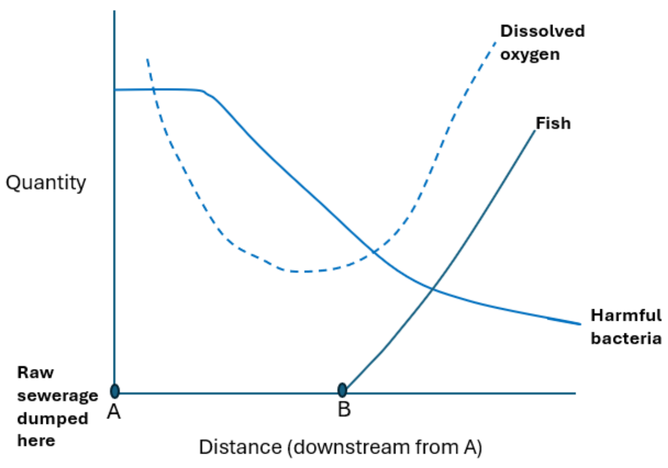
b. Enzyme activity decreases and stops at 40 ^{o}C. The enzyme is denatured and will not function again at 30 ^{o}C.

c. Enzyme activity decreases and stops at 40 ^{o}C. The enzyme is denatured and will function as normal again at 30 ^{o}C.

d. Enzyme activity increases and stops at 40 ^{o}C. The enzyme is slowed and will function again at 30 ^{o}C.

Questions 35 to 38 relate to the following graph.

The graph shows what has happened to the fish in a river after the dumping of raw sewage at point A.



Graph adapted from: Jones & Jones, *Cambridge C-Ordinated Science Biology*, Cambridge University Press

35. Why is the number of bacteria so high at A?

a. The dumped sewage contains lots of oxygen

b. The dumped sewage contains lots of bacteria

c. The dumped sewage contains no fish to eat bacteria

d. Raw sewage was not in fact dumped at A

36. What happens to the number of bacteria as you move downstream from the dumping site?

The bacteria:

a. Spread out so that they are spread evenly in the river

b. Spread out so that they are more concentrated

c. Spread out so that they are less concentrated

d. Spread out and produce oxygen

37. Why did the oxygen in the water decrease?

a. Fish are using the oxygen

b. Bacteria are using the oxygen

c. Oxygen is being swept downstream

d. UV radiation is ionising the oxygen

38. What may have happened at B that has enabled the fish to increase in number?

a. Because there are more bacteria, fewer fish die; oxygen levels are higher, and available for the fish.

b. Because there are less bacteria, fewer fish die; oxygen levels are lower, and available for the fish.

c. Because there are less bacteria, fewer fish die; oxygen levels are higher, and available for the fish.

d. Because there are less bacteria, more fish die; oxygen levels are higher, and available for the fish.

Questions 39 to 42 relate to the following information.

Stomate are defined as any of the microscopic openings or pores in the epidermis of leaves and young stems. Stomata are generally more numerous on the underside of leaves. They provide for the exchange of gases between the outside air and the branched system of interconnecting air canals within the leaf.

39. Plants lose water through stomates, so why would a plant in a hot, dry habitat open its stomates at all?

    a. To attract pollinators.

    b. To allow passage of gases for photosynthesis.

    c. To regulate temperature.

    d. To expel noxious chemicals.

40. Consider two plants in the same habitat. Plant A opens its stomates early in the morning. Plant B opens its stomates at midday. Which plant will lose more water through its stomates? Why?

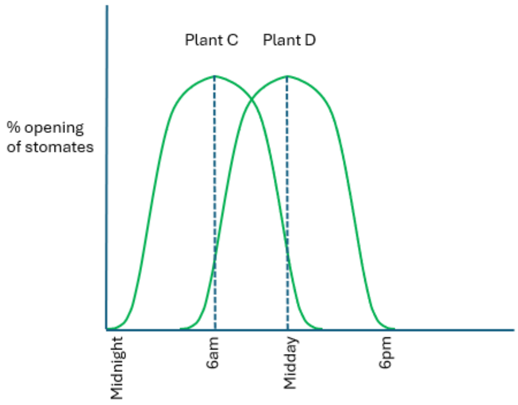
    a. Plant A because temperature and therefore evaporation are higher at midday.

    b. Plant B because temperature and therefore evaporation are lower at midday.

    c. Plant A because temperature and therefore evaporation are lower at midday.

    d. Plant B because temperature and therefore evaporation are higher at midday.

41. Consider the two plants in the graph. They are in different environments. When does each plant have its stomates open completely?



a. C at Midnight; D at 6pm

b. C at Midday; D at 6am

c. C at 6am; D at Midday

d. C at 6am; D at 6pm

42. Which of the two plants would you expect to find in a dry habitat? Why?

a. Plant D, as its stomates are open early in the day, when its cool, and open at midday, when it is hot, and evaporation is low. This helps conserve water.

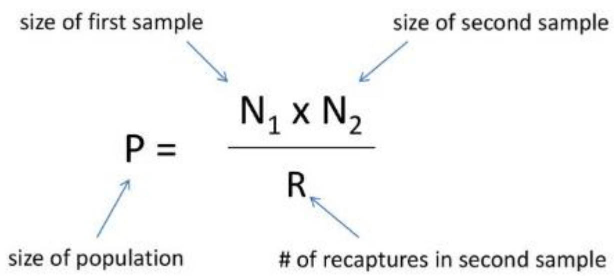
b. Plant C, as its stomates are open at midday, when its cool, and close early in the day, when it is hot, and evaporation is high. This helps conserve water.

c. Plant D, as its stomates are open early in the day, when its cool, and close at midday, when it is hot, and evaporation is high. This helps conserve water.

d. Plant C, as its stomates are open early in the day, when its cool, and close at midday, when it is hot, and evaporation is high. This helps conserve water.

Questions 43 to 45 relate to the following information.

Lincoln Index is a formula used to estimate population size based on mark and recapture. This involves sampling a population, marking them, releasing them back into their natural habitat, then sampling again, counting the number of marked individuals recpatured. The formula is:



After random sampling, 200 snails are captured, tagged, and released. A month later, 150 untagged and 30 tagged snails are captured.

43. Use the Lincoln Index to estimate the total snail population after one month.

a. 12

b. 120

c. 1200

d. 12000

44. Two months later the snail population was monitored. On this occasion 120 untagged and 20 tagged snails were captured.
Did the snail population increase, decrease or remain steady over the two time points?

a. Increased

b. Decreased

c. Remained steady

45. What would the effect on the estimate of marked individuals having a higher mortality than non-marked individuals (i.e. the marked snails were easily identified and eaten by birds)?

a. The population size estimate would be lower than expected

b. The population size estimate would be higher than expected

c. The population size estimate would be the same as expected

d. The population size estimate would be significantly affected by decreased competition

Questions 46 to 49 relate to the following information.

Structures within our bodies often have increased surface area to increase the rate of exchange, as seen in the alveoli in the lungs, which function in gas exchange, and microvilli in the intestinal wall across which nutrients in the gut are absorbed. Structures may also increase exchange efficiency by decreasing their volume, which is a feature also seen in the microvilli. Thus, we can conclude that having a high surface area to a low volume is an adaptation in structures used in exchange or absorption of materials. That is, having a high surface area to low volume ratio is beneficial. This ratio is calculated by total surface area/ by total volume.

The total surface area (mm^2) and total volume (mm^3) of structures 1, 2, 3, and 4 are listed in the table below.

Structure	Surface Area (mm^2)	Volume (mm^3)
1	80	4
2	175	20
3	90	45
4	1,000	200

46. Which of the following options ranks the structures in order of **INCREASING** ability to absorb nutrients well?

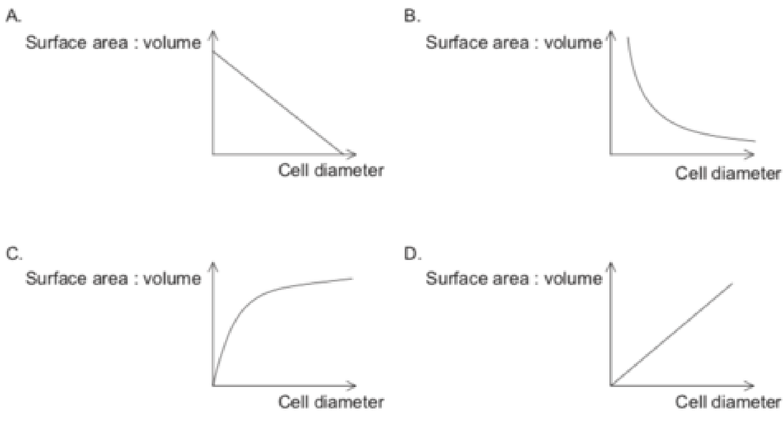
    a. 1 < 3 < 2 < 4

    b. 3 < 4 < 1 < 2

    c. 1 < 2 < 4 < 3

    d. 3 < 4 < 2 < 1

47. Which graph represents the change in cell surface area to volume ratio with increasing cell diameter?



[Source: © International Baccalaureate Organization 2019]

    a. A

    b. B

    c. C

    d. D

48. A researcher studying membrane dynamics fuses two small cells together which results in a single larger cell.

    Which of the following best describes what happens to the cell due to an altered surface area to volume ratio?

a. The cell will have difficulty taking nutrients into the cell but will not have difficult eliminating wastes.

b. The cell will be able to easily take in and eliminate nutrients and waste materials.

c. The cell will have difficulty taking in and eliminating nutrients and waste materials.

d. The cell will be able to readily take in materials but have difficulty in eliminating waste.

49. Which of these organisms will have the smallest surface area to volume ratio?

a. Amoeba

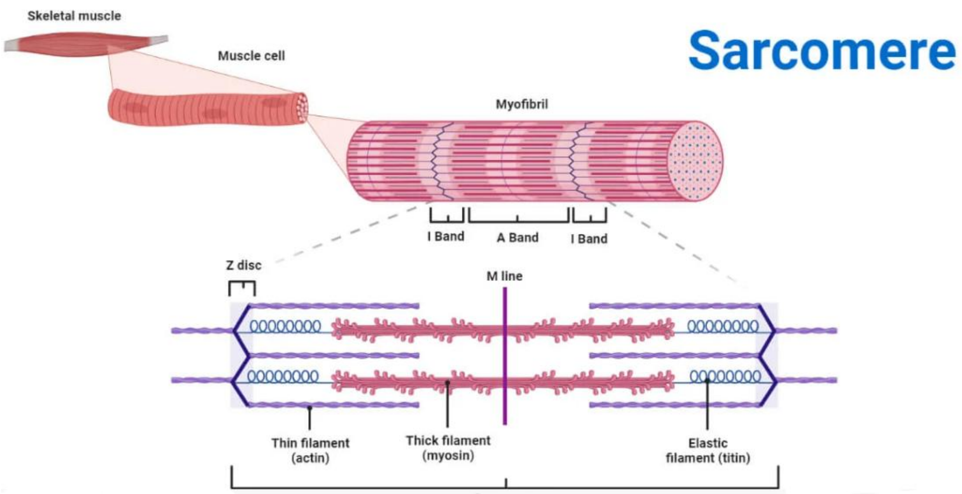
b. Mouse

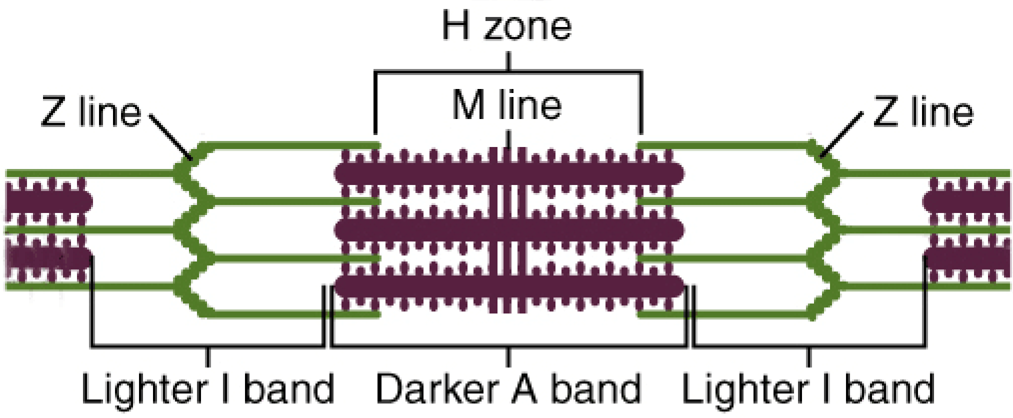
c. Lady Beetle

d. Elephant

Questions 50 to 52 relate to the following information.

Vertebrate skeletal muscle consists *of* contractile units called sarcomeres, which are made up of thin and thick filaments and other associated proteins.





50. When muscles contract the:

a. distance between Z lines becomes greater.

b. H zone, I Band and the distance between Z lines all become smaller.

c. A Band decreases in length.

d. The length of the I Band is not changed.

51. How many sarcomeres are completely visible in the photographed muscle fibre below?

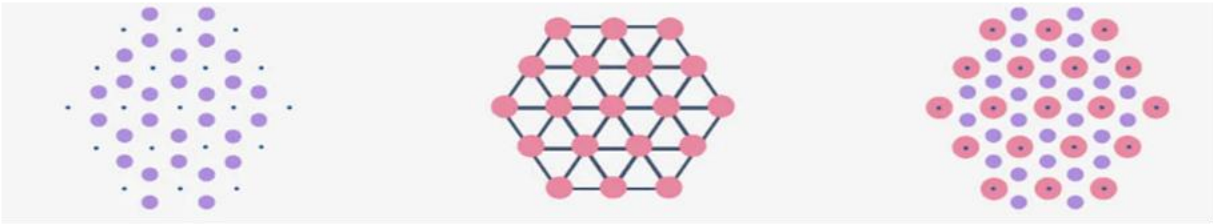
a. 10

b. 12

c. 14

d. 16

52. The below diagram depicts the arrangement of actin and myosin filaments in a sarcomere.

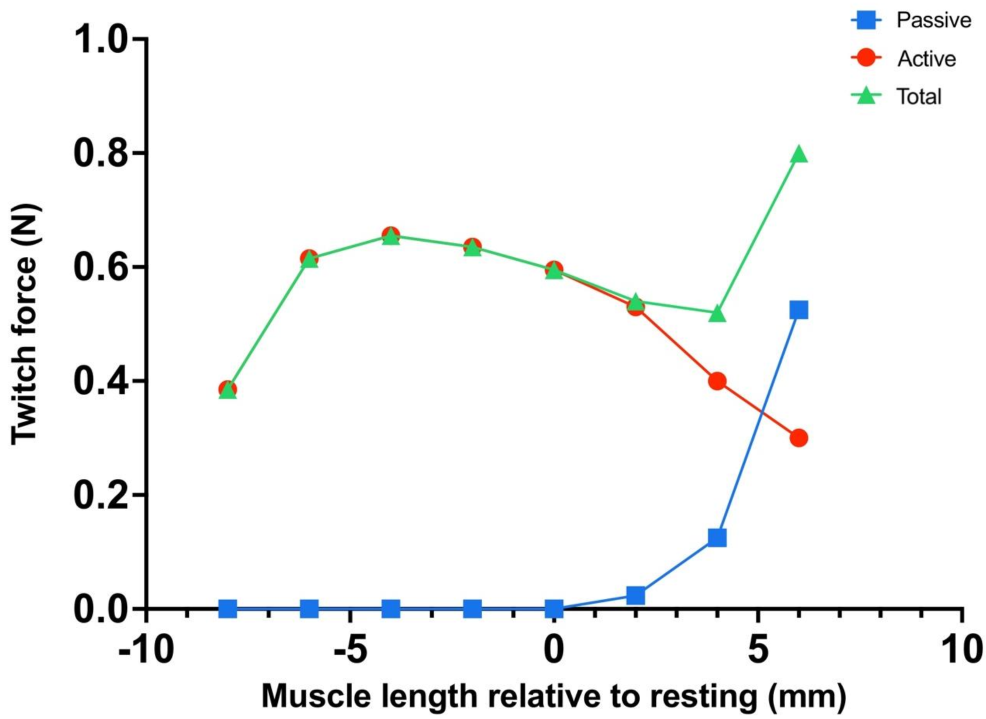


A B C

Match the images A, B and C with the correct zone of a sarcomere from these options.
(H Zone, I Band, overlapping Zone, M Line, Z Line)

Image	Zone
A	
B	
C	

53. The total force generated by sarcomeres is the sum of passive and active force. Active force is created when the fibres of the sarcomere interact and slide past one another, while passive force is generated when elastic elements of the muscle are stretched. In the experiment below, the gastrocnemius muscle of the cane toad *Bufo marinus* was removed and stretched to different lengths and the force that it generated was measured.



Source: https://litfl.com/cardiovascular-physiology-overview/

When the *Bufo marinus* gastrocnemius muscle is stretched to 4mm and contracted, what percentage of total force is generated by passive force?

a. 22%

b. 30%

c. 35%

d. 38%

54. Proteins are an essential part of life, serving countless different roles within every organism. Proteins consist of long chains of amino acids, which are a group of molecules with similar structures. These amino acids are coded for by 4 different DNA bases (Adenine, Cytosine, Guanine and Thymine).
Every unique sequence of 3 bases, which can be any of the four bases, codes for one specific amino acid. For example, the sequence CGC (Cytosine-Guanine-Cytosine) codes for the amino acid Arginine. More than one sequence can code for the same amino acid. In humans, there are 20 different amino acids used to make proteins.

An alien species has been discovered that uses 72 different amino acids, which means they must have at least 72 unique base sequences.

Which of the following is a possible explanation for this?

a. The alien species uses the base uracil instead of thymine.

b. The alien species only uses 3 different bases to code for amino acids.

c. Each alien amino acid is coded for by a sequence of 4 bases instead of 3.

d. Humans also have at least 72 unique base sequences, but most of them code for the same amino acids.

55. In Eukaryotic cells DNA that sits inside the nucleus is expressed when it is transcribed into single strand RNA molecules that are able to pass through the nuclear membrane into the rest of the cell. Then this RNA is translated by ribosomes which make a corresponding sequence of amino acids out of free amino acids that then become functional proteins. A made up single celled eukaryotic organism has the following proteins of note:

Name of Protein	**Role**
ef-1	Synthesis of protein associated with transcription
bh-5	Synthesis of protein associated with correct ribosome structure
kal-1	Synthesis of protein associated with correct outer membrane function
rtr-2	Associated with maintaining nuclear membrane structure

Consider a wild type lineage of this species that has had its DNA randomly mutated in a clinical setting. The cells of these lineages are clones. Some biologists wish to identify which of the genes associated with the encoding of the above proteins, if any, have been affected by the mutation. They conduct some initial tests with results shown below, by comparing the mutant lineage to a wild type lineage.

Test	WT	M1
Cells were exposed to extracellular environment with a compound that stains proteins	Stained	Unstained
Cells were exposed to extracellular environment with compound that stains RNA	Unstained	Unstained
Cells were tested for presence of RNA	Standard result	Standard result
Cells were tested for presence of free amino acids	30 unknown units	572 unknown units

From these results the scientists have some idea as to what mutations might have occurred. Recommend the best choice of a follow up test so that the biologists might be most confident in their conclusion from the results.

a. Expose the cells to many different staining compounds that pass through the cell membrane of the wild type.

b. Expose the intracellular contents to a fluorescent synthetic protein that binds to ef-1.

c. Insert fluorescent amino acids into both lineages and observe long term behaviour.

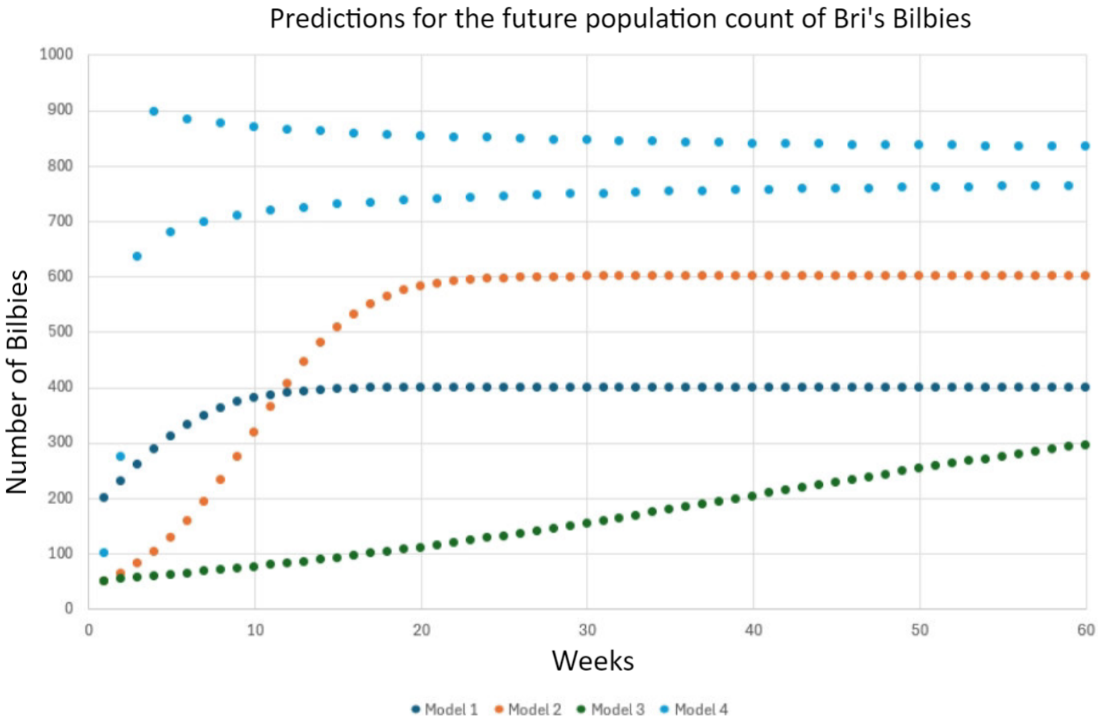
d. Replace the nuclear membrane of a cell from the mutated lineage with that of a cell from the wild type lineage and run the preliminary tests again.

Questions 56 to 63 relate to the following information.

Bri has a population of Bilbies. She wants to create a model to predict the future count of her Bilby population. She read online that the equation:

$$P_n = P_{n-1} + r * P_{n-1} * (1 - \frac{p_{n-1}}{K})$$

Is a popular model with biologists that predicts how large a population will be at one time based on how large it used to be. Where *r* is called the rate of growth and *K* is the carrying capacity. She reads that the rate of growth is how quickly a population will change and the carrying capacity is the maximum number of individuals that can be supported. She then makes 4 different versions of these models using different choices of *r* and *K* with the hope of seeing which one is best.



56. Model 2 predicts more Bilbies than model 1 at week 60 because it has a higher rate of growth

a. True

b. False

57. Model 2 crosses model 1 after approximately 11 weeks because it has a higher carrying capacity but a smaller rate of growth

a. True

b. False

58. Model 4 has the highest rate of growth.

a. True

b. False

59. The reason for Model 4's unusual oscillating behaviour is that it has a higher population at week 4 than its carrying capacity.

a. True

b. False

60. It is harder to tell the carrying capacity of model 3 from the information we are given than for any other model.

a. True

b. False

61. Bri does some tests and finds that the best model at predicting her Bilby population count has parameters $r = 0.7$ and $K = 500$. She finds that when she does calculations with her model there is a population count x such that when she uses her model to predict how many Bilbies there will be next week she gets the same number, that is x again.

Give the value for x to the nearest whole number.

a. 200

b. 250

c. 500

d. 750

62. The Fibonacci sequence is a list of numbers where the next number is calculated from the previous one like in our population model. The Fibonacci sequence starts with 1 and keeps getting larger without stopping (1,1,2,3,5,8,13,21…). It was originally observed by Fibonacci that we could model the population of rabbits on an island using a variation of this sequence called the Fibonacci model given by:

$$P_n = P_{n-1} + P_{n-2}$$

On one hand the Fibonacci model is good because it reflects the fact that a population should get larger at an increasing rate. However, we do not actually use the Fibonacci model to predict population growth.

Select the valid reason as to why the Fibonacci model is not a good model of population growth:

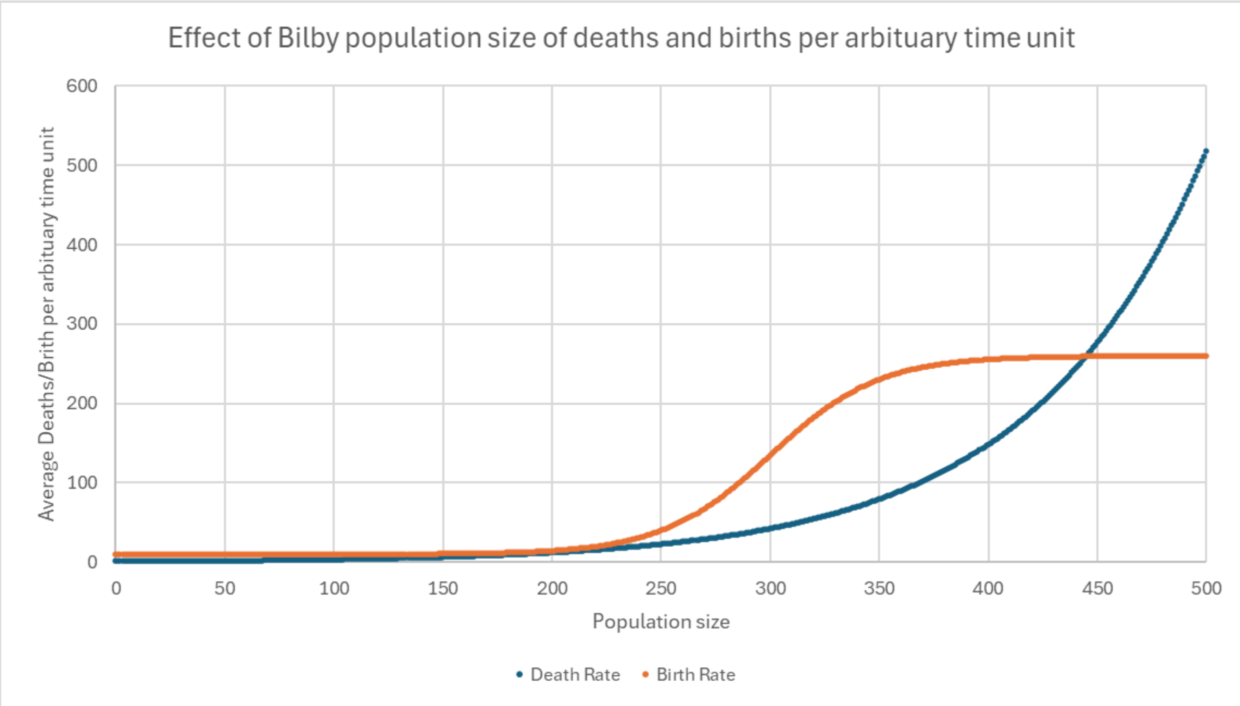
a. The population of a species cannot keep increasing forever and must eventually stop increasing.

b. The Fibonacci model cannot be modified to reflect that some species have different times between generations.

c. The Fibonacci model requires that the population start at 1 which means that for any sexually reproducing species there cannot be any growth of the population.

d. The Fibonacci model is too volatile.

63. Deciding to stick with the model she had at the start Bri finds another population of Bilbies and plots the birth and death rate as so:



Give the carrying capacity Bri should use in her model to the nearest whole number:

a. 305

b. 425

c. 445

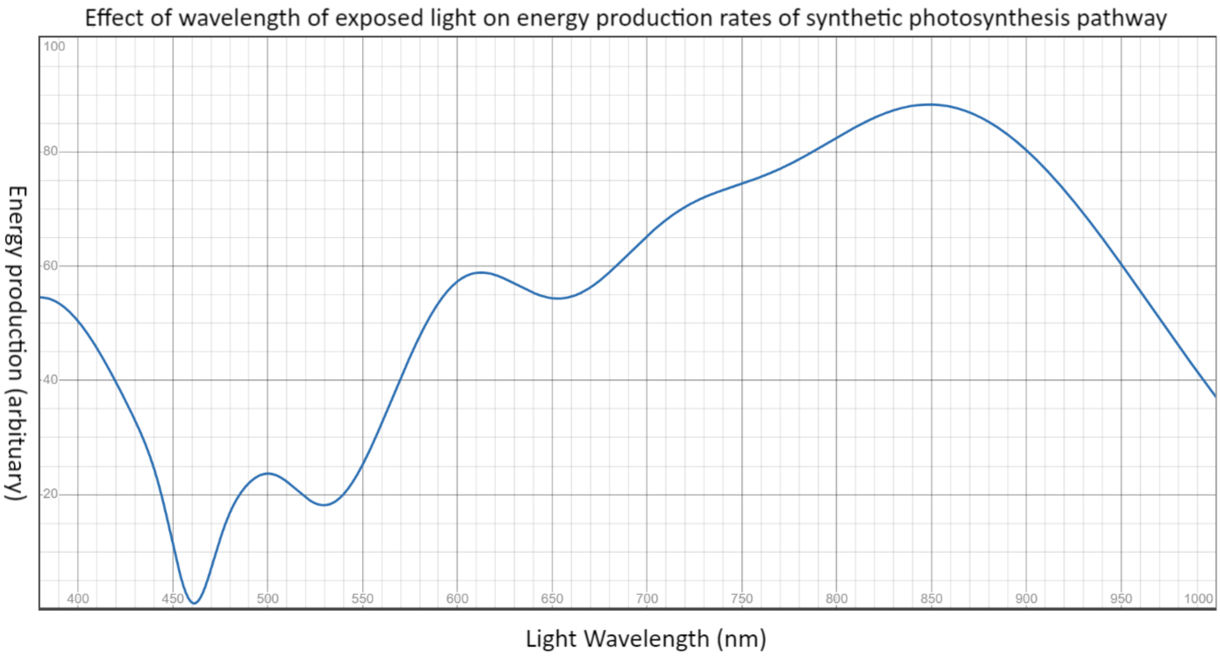
d. 500

64. Plants are green because they contain the pigment called chlorophyll which absorbs light energy and uses this energy to create glucose in photosynthesis. Chlorophyll is green because it reflects green light into our eyes. A scientist decides that she will create her own synthetic photosynthesis pathway complete with a pigment she calls "pigment X" to absorb sunlight. Below is a graph of some results from experiments she has run on her synthetic photosynthesis pathway that has been made to appear continuous and a figure labelled '1' that matches colour of light to its wavelength.

Use the following to answer the question.

Figure 1:

Colour	Wavelength
Violet	380-450 nm
Blue	450-485 nm
Cyan	485-500 nm
Green	500-565 nm
Yellow	565-590 nm
Orange	590-625 nm
Red	625-750 nm
Infra-Red	750 nm - 1mm



What colour is the pigment she used?

a. Blue

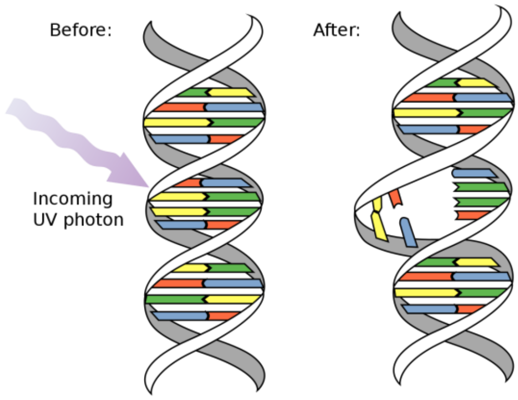
b. Green

c. Orange

d. Infra-red

Questions 65 and 66 relate to the following information.

UV light is a non-visible wavelength of light that is a component of sunlight. UV light is considered a carcinogen, that is, it causes cancer by mutating DNA. DNA consists of two strands that are a sequence of bases. The bases can be either A, T, C or G. The stands are attached by bonds between these bases. A and G are called purines and T and C are called pyrimidines. Bonds form either between A and T or C and G. So, if we know the sequence of bases in one strand, we know the sequence of bases in the other. U is found in RNA instead of T, it also bonds with A and is also a pyrimidine. UV light causes a mutation in DNA or RNA as if there are two pyrimidines in a row in a single strand of DNA or RNA, UV light may bond them together in a strange way that forms a kink in the strand that prevents it from attaching to the other strand properly, we call this a pyrimidine dimer. This can happen between any two pyrimidines, but it is far more likely to happen between two Ts.



https://en.wikipedia.org/wiki/Pyrimidine_dimer#/media/File:DNA_UV_mutation.svg

65. Suppose the following pieces of RNA and DNA are exposed to the same amount of UV light.

Which is most likely to have the most pyrimidine dimers? (You should assume the actual sequence of the RNA and DNA is in a random order)

a. 100 base pairs (bp) of RNA

b. 50 bp of DNA molecule with equal distribution of all bases

c. 100 bp of DNA molecule containing 5% Guanine bases

d. 100 bp of DNA molecule containing 40% Guanine bases

66. A key mechanism to repair these dimers is called photoreactivation, A protein called photolyase uses energy captured from visible light to break these dimers back into the two original bases.

An experiment is conducted with Wild type *E. coli* cultures. Two independent variables are being tested for their effect on the survival rate of *E. coli* after being exposed to UV light. The independent variables are: The length of exposure to UV light that *E. coli* are subjected to and whether or not the *E. coli* are exposed to visible light when they are not being exposed to UV light.

The scientists set up 6 groups.

Visible light, no UV exposure	Visible light, 1s UV exposure	Visible light, 5s UV exposure
No visible light, no UV exposure	No visible light, 1s UV exposure	No visible light, 5s UV exposure

Which of the following are likely to be observed?

a. Increasing the length of UV exposure will decrease the survival rate; Working with *E. coli* in complete darkness will have no effect on the survival rate.

b. Increasing the length of UV exposure will decrease the survival rate; Working with E. coli in complete darkness will decrease the survival rate.

c. Increasing the length of UV exposure will not have a significant effect on the survival rate; Working with *E. coli* in complete darkness will increase the survival rate.

d. Increasing the length of UV exposure will not have a significant effect on the survival rate; Working with *E. coli* in complete darkness will decrease the survival rate.

Questions 67 and 68 relate to the following information.

Consider a small region of DNA that is only 15 base pairs long. Below is the sequence of the coding strand and the template strand.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Coding strand	G	G	C	A	A	C	A	A	T	G	G	G	A	A	C
Template strand	C	C	G	T	T	G	T	T	A	C	C	C	T	T	G

Bioinformatics is a field of biology that analyses (often much larger) strands of DNA by applying mathematical tests to the DNA. One such test is to calculate the GC skew of a given strand. This is calculated by counting how many bases labelled G occur and how many bases labelled

C occur. These are labelled as *g* and *c* respectively. Then we calculate the GC skew with this formula:

$$\text{GC Skew} = \frac{g - c}{c + g}$$

The reason we use this is because if our DNA bases were random the number of bases labelled as G and as C should be very similar. So, the value of *g* should be roughly equal to *c*. Then *g* - *c* should be very close to zero and according to the formula so should the GC skew.

Therefore, if the GC skew of a strand of DNA is not close to zero this tells us that the strand of DNA has a very specialised role in the organism it came from.

67. Calculate the GC skew of the coding strand above to 2 decimal places.

 a. 0.25

 b. 0.50

 c. 0.75

 d. 1.00

68. A scientist uses a computer program to find the GC skew of the coding strand of a region of human DNA that is about 10 million bases long. The program returns a value of -1.4.

 The scientist is sure that the program has returned an incorrect result. Why might this be?

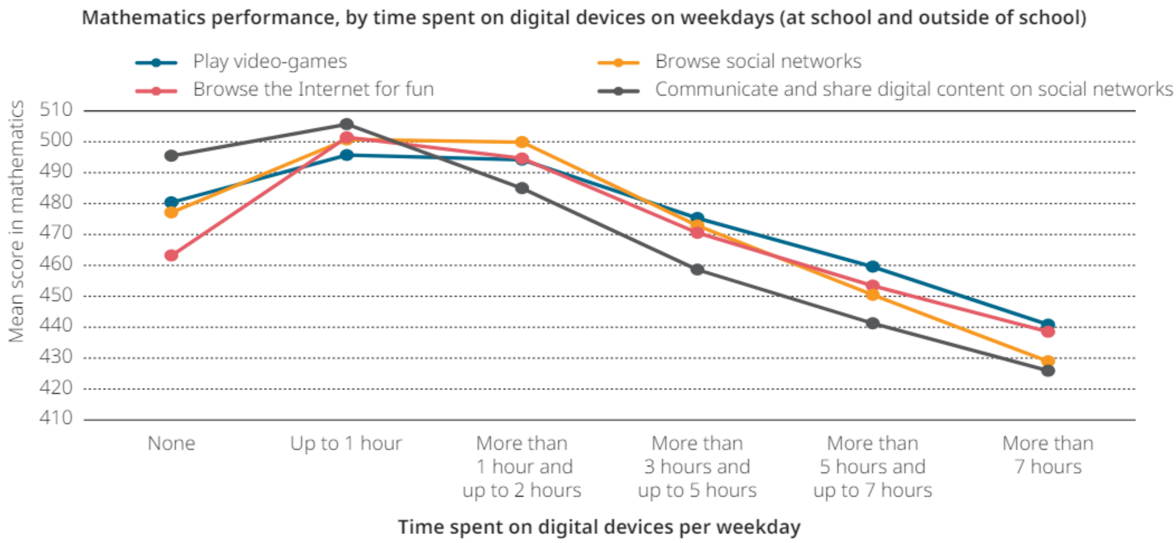
 a. The GC skew of a strand this long should be roughly 0.

 b. It should not be possible to have a negative GC skew.

 c. The GC skew should be between than -0.5 and 0.5.

 d. The magnitude of the GC skew should not be more than 1.

69. The increasing use of digital technologies and devices has raised questions about their impact on the health and development of children. How students use digital resources, and the types of devices they rely on, shape their susceptibility to distractions while using digital technology. The graph below shows the impact of the time spent on digital devices per weekday on students' mathematical performance.



Source: OECD.org

The graph shows that students who play video games:

a. For more than 5 hours and up to 7 hours score higher in mathematics than their peers who play for only 1 hour.

b. For more than 5 hours and up to 7 hours score just as well as their peers who communicate and share digital content on social networks for the same amount of time.

c. Regardless of the digital device being used, students who spend more than 1 hour per day using the device, score lower in mathematics.

d. Have greater mathematical literacy than those students who do not engage with digital devices, regardless of the time spent at school, or outside of school.

70. The below food web shows all the possible energy paths in a particular marine environment.

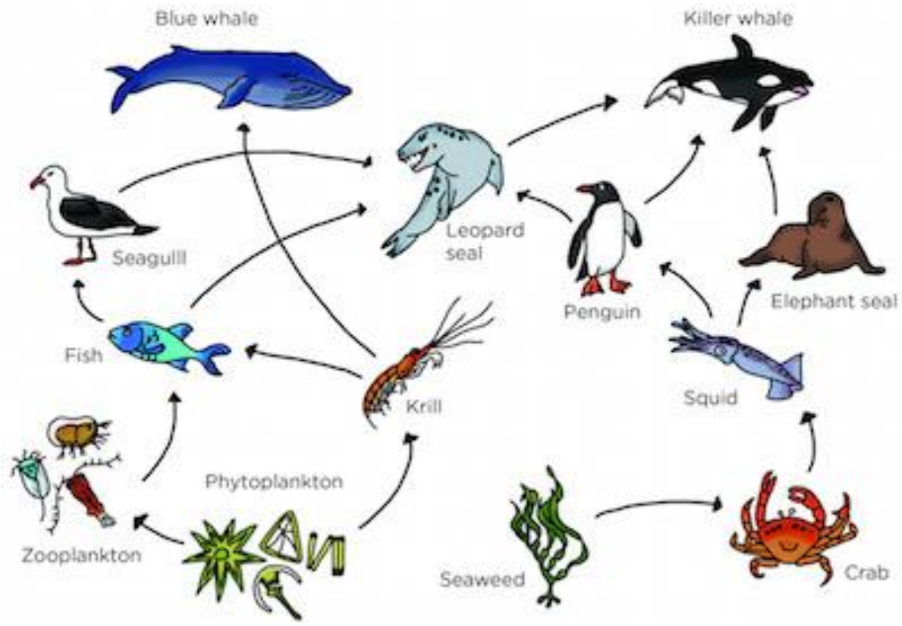


Image sourced: www.complexityexplorer.org.

Which of the below statements is most correct?

a. The Blue whale and Killer whale are examples of primary consumers

b. Phytoplankton and Seaweed are secondary consumers

c. Phytoplankton and Seaweed are examples of the 1st trophic level in a food chain

d. Seagulls and Leopard seals are examples of secondary consumers

END OF EXAM

We would like to acknowledge the contribution and inspiration to this paper by the New Zealand International Biology Olympiad

2024 ASOE Biology - Answers

Question	Answer	Question	Answer	Question	Answer
1	b	31	c	59	b
2	d	32	b	60	a
3	b	33	a	61	c
4	b	34	b	62	a
5	a	35	b	63	c
6	d	36	c	64	a
7	c	37	b	65	c
8	b	38	c	66	b
9	b	39	b	67	a
10	c	40	d	68	d
11	c	41	c	69	c
12	c	42	d	70	c
13	b	43	c		
14	d	44	a		
15	c	45	b		
16	a	46	d		
17	b	47	b		
18	c	48	c		
19	b	49	d		
20	a	50	b		
21	b	51	d		
22	c	52 a	I Band		
23	c	52 b	M Line		
24	c	52 c	Overlapping zone		
25	d	53	a		
26	a	54	c		
27	b	55	d		
28	a	56	b		
29	c	57	b		
30	c	58	a		